BEST STRATEGIES TO REDUCE LIGHT POLLUTION



Image source: darksky.org

Humans are a diurnal(daytime) species. This means that our eyes evolved to function best in bright light during the day. However, we still need lighting to be able to do certain day-to-day activities during the night and to secure our living spaces. Well-planned outdoor lighting provides visibility, allowing us to carry out many important tasks during the night with ease.

However, improperly designed lighting can disrupt both human life and wildlife as well. All animals need different level of daylight and darkness. But one thing for sure, human disrupted their circadian rhythm by turning on poorly designed artificial lighting.

What is light pollution?

Light pollution is defined as **"the inappropriate or excessive use of artificial light"** by the International Dark-Sky Association (IDA). It is caused primarily by artificial light from electric lights at night which scatter in all directions away from their source instead of being confined to just where they are needed.

It is exacerbated by excessive, misdirected, or obtrusive use of light, and has the ability to alter natural conditions, compromising human health and disrupting ecosystems.

This stray light causes glare and reduces visibility on roads, interferes with astronomical observations including stargazing, disrupts ecosystems like migratory bird routes that depend on natural levels of ambient light at night, changes our perception of stars, and has adverse effects on nocturnal animals like frogs or insects who respond to moonlight.

What Is Glare Light Pollution?

The term "glare light pollution" refers to excessive brightness. You may associate glare with squinting and other visual discomfort felt in the presence of artificial light, such as bright street lamps.

What Is Skyglow Light Pollution?

Skyglow light pollution can be seen from satellites, airplanes, and even the International Space Station. Skyglow is the brighter night sky seen over people's homes.

What Is Light Trespass?

Light trespass is the term used to describe stray light that falls outside of its intended area.

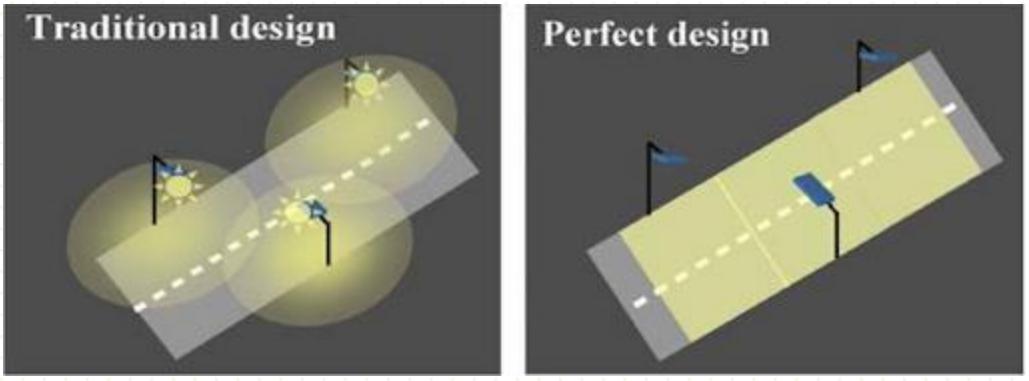


Image source: zdnet.com

What Is Clutter Light Pollution?

Clutter type light pollution is common in urban areas, where multiple lights may overlap, resulting in an overly bright and confusing light environment.



Image source: assa.org.au

While light pollution may not initially appear to be a big deal, it is a big deal for our health. Light pollution makes a significant contribution to chronic environmental and health problems for every affected individual, animal, and plant, according to a National Geographic.

WHAT ARE THE WAYS LIGHT POLLUTION HURTS OUR PLANET?

Light pollution is a pervasive and serious problem that has been going on for decades. When too much light enters the night sky, it disrupts natural circadian rhythms of humans sleeping as well as all other forms of life in darkness such as plants or wildlife.

Devastates wild life



Image source: herpsafari.com

Plants and animals rely on the daily light and dark cycle of the Earth to guide their life-sustaining behaviors. According to research, artificial light at night has negative, even fatal, effects on many species.

Light pollution has been shown to affect the natural behavior of animals. Nocturnal predators have also adapted their hunting habits according to human lighting, depriving prey species such as insects and frogs from critical feeding opportunities during nighttime hours.

Harms your health



Image source: koruhastanesi.com

When we sleep, our bodies produce melatonin, which helps regulate things like body temperature, metabolism, and heart rate, among other things, to keep us healthy while we sleep.

However, if we are exposed to artificial lights at night, this production process can slow down, causing us not to fall asleep quickly and not get rested well enough from our sleep.

According to a research, artificial light at night has a negative impact on human health by increasing our risk of obesity, sleep disorders, depression, diabetes, and other diseases.

Reduces safety



Image source: interior-news.com

Excessive outdoor lighting can cause a number of problems. It reduces visibility, and when paired with an ineffective design it also decreases the safety of places lighted by these lights.

There is no conclusive scientific evidence that more outdoor lighting deters crime. In fact, poor outdoor lighting can reduce personal safety by increasing the visibility of victims and property to criminals.

Wastes energy & resources

According to IDA, at least 30% of all outdoor lighting in the United States is wasted, primarily by lights that aren't shielded causing an increase in greenhouse gas emissions, contributing to climate change, and making us all more energy dependent.

HOW DOES LEED V4 DEAL WITH LIGHT POLLUTION?

The LEED rating system, particularly LEED v4 O&M Existing Buildings, does not mandate light pollution reduction as a requirement for projects to abide by. However, it is a credit that is relatively easy to achieve if the lighting system is designed using certain best practices. Projects planning to comply with the credit are given two options from which to choose.

Fixture Shielding

LEED v4 O&M requires shielding all exterior fixtures where the sum of the mean lamp lumens exceeds 2,500 such that any light emitted at a vertical angle does not exceed 90 degrees from straight down.

Projects teams can acquire lamp lumens information in the manufacturer's technical data sheets. If this information is not available in there, photos of the luminaires might be used in conjunction with other supporting information to look for credit documentation. The goal is to determine fixtures that emit high amounts of light at a vertical angle that can be shielded.

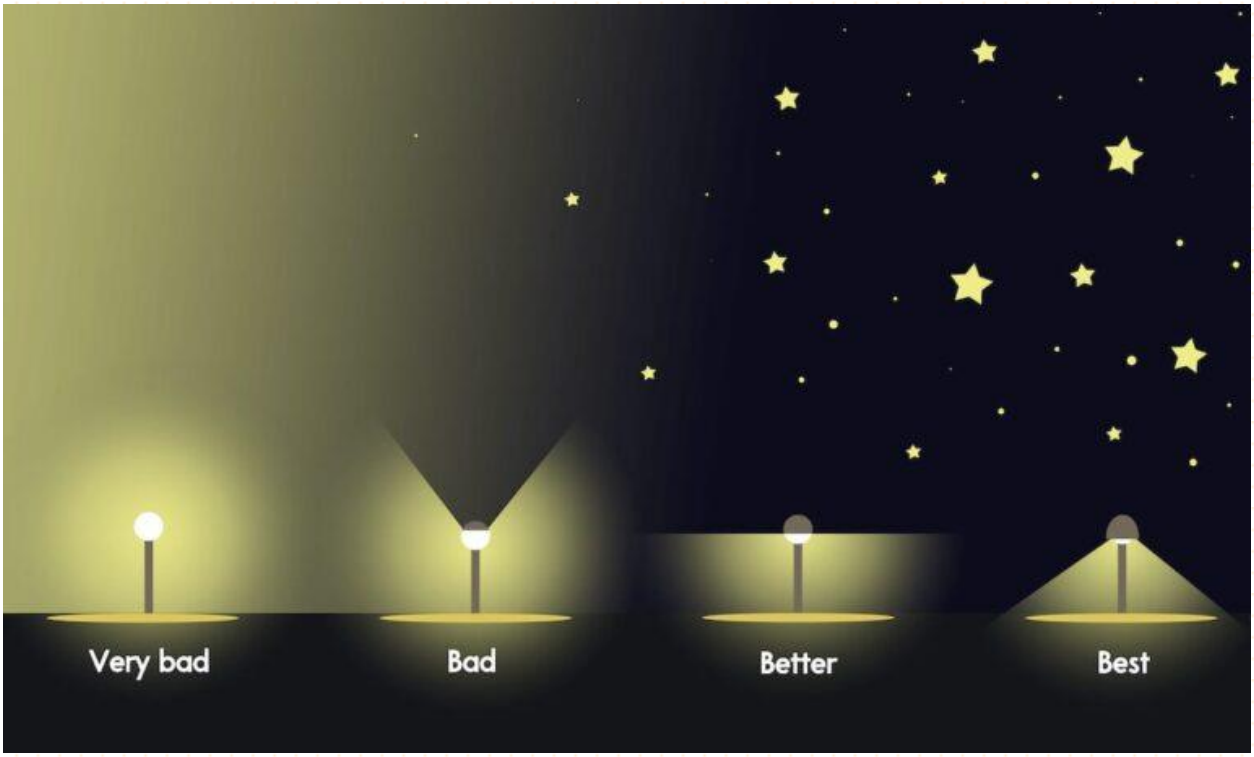


Image source: gostargazing.co.uk

Once the noncompliant luminaries have been identified, the project team should investigate options for retrofitting or replacing noncompliant existing site lighting to meet the credit requirements. Shielding to retrofit existing noncompliant luminaires may be provided from the manufacturer or may need custom-design work. It is important to attach shielding to the fixture properly, else leaking light can significantly reduce efficiency, causing both light pollution, and safety issues.

Another option project teams can use is swapping out the lamps, instead of shielding retrofits. One thing worthy mention, noncompliant luminaires with a total of 2,500 or less mean lamp lumens will still meet the credit requirements since they are not considered large enough through to contribute light pollution. Shielding fixtures is usually an easy task that can be done by a qualified electrician. Existing building projects can quickly implement shielding for their projects and earn one point rewarded to comply with this credit's requirements.

One great tip that would further reduce light pollution is installing low-hanging bulbs, or keeping the height of lighting fixture low to further reduce light pollution. Since higher the lighting fixture more problem artificial lighting tends to create.

Perimeter Measurements

Another option offered for projects is taking perimeter measurements for the night illumination levels at regularly spaced points on the project boundary. This option is more result-oriented, rather than the fixture shielding option that we have discussed.



Image source: houzz.in

If a building project does not significantly contribute to light pollution, documenting this fact allows the project to be compliant with the credit.

Light pollution credit requires at least eight measurements to be done with a maximum spacing of 100 feet (30 meters) in between each measurement point. And the illumination level measured with the lights turned on shouldn’t be more than 20% higher than the level measured with the lights turned off.

This is a very objective goal that can be very useful for projects that don't have much exterior lighting or use another technology that allows them not to contribute to light pollution but not intend of installing shielding.

Light pollution reduction credit doesn't have any performance targets to achieve. So, a project can be either compliant with the requirement and earn the point, or, be incompliant and earns no point through this credit.

Project teams should begin by identifying the points along the project boundary where illumination levels can be measured and compared to the credit requirements. The team should ensure that there are at least eight measurement points, with a maximum distance of 100 feet (30 meters) between points, and that the measurement points are equidistant to fairly represent the lighting levels at all points along the project boundary.

The team takes illumination measurements with the lights turned off at each measurement point along the project boundary. Then it repeats the measurements at the same points with the exterior and site lights on. Any light meter capable of measuring illumination at ground level can be used to take measurements.

To make consistent measurements, the measurements must be taken on the same day, either after civil twilight in the evening or before civil twilight in the morning. Twilight times can be found online for any location (globally).

Civil twilight is the period of time in the morning (dawn) or evening (dusk) when the sun's center is 6 degrees below the horizon. Civil twilight is the best time to distinguish terrestrial objects in good weather conditions. Artificial illumination is normally required to carry out ordinary outdoor activities before civil twilight in the morning and after civil twilight in the evening.

When making outdoor lighting level measurements the interior lights of the building must remain in the same state (on or off). Once the measurements are done, project team compares measurement results to look for compliance with the credit requirement. The illumination measurements collected across the project boundary are not allowed to be averaged, and each measurement point is expected to be compliant by itself.

WHAT CHANGES EXIST IN LEED V4 VERSION OF CREDIT DIFFERENT THAN LEED V3 OF THE CREDIT?

LEED v4 Light Pollution Reduction credit no longer has any interior lighting requirements as it was in v3, it is now solely focused on exterior lighting. Also, the previously available methodology for perimeter measurements to establish compliance by reaching maximum footcandle (lux) levels, as stipulated by IESNA RP-33, is no longer available within the v4 as it was in v3.

WHAT ADDITIONAL STRATEGIES CAN HELP IN REDUCING LIGHT POLLUTION?

Use LED Lights

LEDs and compact fluorescent lights (CFLs) can help save energy and protect the environment. LED lighting allows for lower luminance without sacrificing visibility. Only warm light sources are recommended by IDA for outdoor lighting.



Image source: lightbulbs-direct.com

Projects should look for replacing all glaring outdoor lights with low-glare alternatives. There are numerous types of sky-friendly outdoor lighting alternatives on the market. So it is best to invest time in research and funds to choose one that doesn't contribute to lighting pollution.

Use motion sensors on outdoor lights.

Instead of leaving lights on all night for security reasons, installing motion sensors on the lights can be beneficial. Motion sensors will turn on the lights only when motion is detected, lowering overall electricity costs while also reducing nighttime light pollution.



Image source: quora.com

Dimmers and timers can also assist in lowering average lighting levels and saving even more energy. Outdoor fixture lights can be set to turn off after a certain amount of time, or they can work by gradually turning down the brightness over a specific period of time.

There are different technologies available for those who want to reduce their energy consumption while also lowering light pollution. Lighting upgrades can be done easily at a low cost or an even lower upfront expense with rebates and tax incentives offered by most utility companies, which in turn bring down the monthly payment as well.

Avoid blue lights

Blue-rich white light sources are also known to cause glare and impair human vision, particularly in the aging eye. These lights pose potential road safety issues for both motorists and pedestrians.



Image source: psychcentral.com

According to experts at Harvard Medical School, blue light at night has been shown in natural settings to negatively affect wildlife behavior and reproduction, particularly in cities, which are frequently stopover points for migratory species.

Outdoor lighting with a high blue content is more likely to exacerbate skyglow because it has a much larger geographic reach than lighting with lower blue content. Light pollution can be reduced by using dark-sky friendly lighting, turning lights off when not needed, and minimizing the use of bright or blue lights that contribute to skyglow (the brightening effect that makes it difficult to see stars from large distances).

This will allow for a better night sky, improved visibility at night, and reduce the consequences of development on wildlife and people. The solution is to turn down our outdoor lighting in order to create deeper nighttime darkness so we might finally have some peace under the moonlight.

Use glow stones for all of your outdoor night lightings.

Glow stones emit only 5-7 candelas of light and do not 'cast' light in the same way that electric lights do. The ambient glow from glow stones does not contribute to bright skies at night and is not a source of light pollution. Instead of using bright electric lights, use glow stones to line pathways, steps, and other outdoor areas.



Image source: pinterest.com

Even better, these types of stones do not break and do not require any technical knowledge to use.

In fact, these stones can be used to detect light pollution. If you can see the glow at night, it means you live in a high-light-pollution area. If the glow is faint or not visible, you are in an area with not much of light pollution.

Keep light trespassing to a minimum.

Bright outdoor lights directed at neighboring houses are completely unethical because they cause discomfort. As a result, it must be ensured that outdoor lights do not intrude into residential properties.



Image source: pinterest.com

It is as simple as making sure that the lighting in your home or street lighting projects does not allow light to enter neighboring homes. It is vexing and has even been linked to the development of sleep disorders. In short, avoid using outdoor lights that cast a glare into your neighbor's window.

Purchase only IDA-approved light fixtures

The International Dark Sky Association certifies dark sky-friendly lighting fixtures that adhere to their stringent guidelines. When shopping for new lights, look for this symbol:



Image source: darksky.org

The Fixture Seal of Approval is an objective, third-party certification for luminaires that reduce glare, light trespass, and do not pollute the night sky. You can also help by utilizing energy-saving features such as timers, motion sensors, and dimmers; ensuring that your light fixtures are shielded so that light shines down rather than up; and installing lights only when and where they are required.

Set a good example by turning off your lights.

Turning off your lights is the simplest way to help reduce light pollution. This not only helps to reduce light pollution, but it also lowers your energy bill and carbon footprint, while also revealing the beauty of our world in the dark.

Colored lights can be a good alternative

Colored lights are anti-glare and still provide adequate night lighting. Because they do not interfere with night vision, yellow, red, and amber lights can help to mitigate the negative effects of lighting at night.



Image source: architizer.com

REFERENCES

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2 – LEED v4 O+M: Existing Buildings, Sustainable Sites, Light pollution reduction

3 – Light Pollution Solutions, IDA, International Dark Sky Association

4 - The Dark Side of Light: Effects of Light Pollution on Insects and Agriculture

5 - Blue light has a dark side